

Bulk one-defect, refuted as rank-one

The threshold frame and the C_m -phase screen after the r399 circularity verdict

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Abstract

After energy decay of the centred mass difference was refuted (r399) and the 3×3 edge chart was written exactly (r401), the remaining bulk statement on the selected sequence is $\text{ind}_-(A_0) \leq 1$ with $A_0 = R_{N-3} - \frac{1}{2}I$. Two honest candidate forms are adjudicated. Form T: a source-defined rank-one frame $\langle x, A_0 x \rangle \geq c\|x\|^2 - C|\ell(x)|^2$ with $c > 0$, equivalently A_0 a positive matrix minus rank one. Form P: a phase statistic of the r399 Dirichlet coefficients C_m that predicts ind_- rather than the energy E . What is proved: the frame implication, rank- r interlacing, and Weyl's inertia bound (finite linear algebra). What is refuted: both candidate forms as routes to P1. The Bernstein–Szegő dual on the holes already has ~ 49 negatives; the prime source lifts 48 of them (a high-rank positive repair, numerical rank 49 on that subspace). Rank one can kill at most one negative. The genuine negative mode of A_0 , when present, is orthogonal to the centred difference (cosine 0.0075). Consecutive C_m -signs do not separate $\text{ind}_- = 1$ from $\text{ind}_- = 21$. Dead χ satisfy P1: they die at $\text{sch} > 0$ (r401), not in the bulk. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Propositions 1 and 2 are finite machine censuses. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The object

Write $R = R_{N-3}$ for the dual Christoffel–Darboux Gram of the first $N_w - 3$ dual OP columns on the hole nodes Y , and $A_0 = R - \frac{1}{2}I$. P1 is $\text{ind}_- A_0 \leq 1$, equivalently $\lambda_2(R) \geq \frac{1}{2}$. Round 398 found a macroscopic cluster of R at $\frac{1}{2}$ from above (FRAME-A: 36 eigenvalues in $[\frac{1}{2}, 3/5]$). Round 399 gave an exact Dirichlet representation of the centred difference d after Fejér, with energy E at quadratic-mean, growing, and with any bound forcing $E \rightarrow 0$ typed RH-near. The reviewer remainder after that circularity verdict is whether P1 still follows from a *threshold* (not a decay) or from the *phases* of the coefficients C_m .

The Bernstein–Szegő reference R^{ref} is the same dual CD Gram built from full-grid Fejér weights on the cosine lattice (r390: $\gamma_k = 1/4$ at residual $2 \cdot 10^{-6}$), restricted to the same Y . The perturbation is $B = R - R^{\text{ref}}$. (An occupied-Fejér dual with a tiny floor on ν -nodes saturates $R^{\text{ref}} = I$ because the dual chain depth $N_w - 3$ exceeds $|Y|$; that degeneracy is disclosed and is *not* the r390 object.)

2 Finite algebra

Lemma 1 (frame implies P1). *Let A be real symmetric $n \times n$. If there exist $c > 0$, $C \in \mathbb{R}$ and a linear form ℓ such that*

$$\langle x, Ax \rangle \geq c\|x\|^2 - C|\ell(x)|^2 \quad \text{for all } x,$$

equivalently $A + C \ell \otimes \ell \succeq cI$, then $\text{ind}_- A \leq 1$.

Proof. $A + C \ell \otimes \ell$ is positive definite, so A differs from a positive-definite matrix by a rank-one update. Cauchy interlacing (or Haynsworth on a 1×1 border) permits at most one negative eigenvalue. $\square$

Lemma 2 (rank- r interlacing). *A symmetric rank- r update changes ind_- by at most r . In particular a rank-one update kills or creates at most one negative eigenvalue.*

Over $\mathbb{Q}$: $\text{diag}(-1, -1) + e_0 e_0^T = \text{diag}(0, -1)$ kills exactly one negative.

Lemma 3 (Weyl). $\text{ind}_-(A + B) \leq \text{ind}_- A + \text{ind}_- B$.

On FRAME-A the Weyl bound through R^{ref} is $49 + 7 = 56$, useless as a proof of P1.

3 Form T, refuted as rank one

Proposition 1 (rank anatomy). *On FRAME-A the Bernstein–Szegő dual on Y has $\text{ind}_-(R^{\text{ref}} - \frac{1}{2}I) = 49$. MAIN kills 48 of those negatives ($\text{ind}_- A_0 = 1$). The lift $M = P_-^{\text{ref}} B P_-^{\text{ref}}$ on the 49-dimensional negative space of the reference has effective rank $r_{\text{eff}} = \|M\|_F^2 / \|M\|_{\text{op}}^2 = 9.55$, leading-share 0.105, and numerical rank 49. The spectral-free surrogate (whole B) has $r_{\text{eff}} = 14.91$. Core-42 has $\text{ind}_- A_0 \in \{0, 1\}$ on 42/42 (28 P1 and 14 vacuous), with $\lambda_2(R) - \frac{1}{2}$ in $[4.3 \cdot 10^{-8}, 2.0 \cdot 10^{-4}]$.*

Lemma 2 then forbids a rank-one B : killing 48 negatives requires rank at least 48. The 36-mode $\frac{1}{2}$ -cluster of MAIN is the remnant of that high-rank lift, sitting razor-thin above $\frac{1}{2}$. A source-defined ℓ built from the centred difference sampled on Y is orthogonal to the genuine negative mode (cosine 0.0075); the frame of Lemma 1 fails for every C in a sealed scan, while the tautological eigenvector works. The distinguished pair (r367) carries only mass 0.151 of that mode: the defect is rest-hosted, not an edge functional on two nodes.

The exact threshold that implies P1, once R^{ref} is *not* already P1, is not a bound on a sub-band energy of C_m . Low-band energy of C_m is already ~ 0 on MAIN, and d is orthogonal to v_- , so Montgomery–Vaughan on short bands—unconditional, and sharp on short intervals—does not see P1. Proving $\lambda_2(R) \geq \frac{1}{2}$ by energy methods is the same wall as r399, now with a named orthogonality.

4 Form P, refuted as a predictor

Proposition 2 (phase screen). *Consecutive-sign correlation of C_m does not track ind_- . FRAME-A has $\rho_1 = -0.341$ and $\text{ind}_- = 1$; scramble (seed 1) has $\rho_1 = -0.363$ (more anti-correlated) and $\text{ind}_- = 21$; energies 1.81 versus 3.39 (factor < 2). Dead χ_{3-15} has $\rho_1 = -0.333$ and $\text{ind}_- = 1$. Flip rate, lag-2, HHI and dyadic band shares likewise fail to separate the two inertias.*

A source-side bound on Weyl phases of $\sum \Lambda(n) n^{-1/2-it}$ is therefore not a route to P1: even a theorem-grade phase law would control the wrong statistic. The honesty gate is not reached, because the empirical predictor is already dead.

5 Named kills

Scramble: $\text{ind}_- = 21$, reference negatives 35, only 14 killed, $r_{\text{min}} = 0.0036$ (the cluster punches through $\frac{1}{2}$). The perturbation is not rank one and does not finish the lift. Two-period ($S = 21$):

$\text{ind}_- = 4$, killed 0 (no Bernstein–Szegő lift). Mutants: omit ℓ (Lemma 1 hypothesis fails); threshold $\times 2$ asks $\lambda_2(R) \geq 1$, false at $\lambda_2 = 0.50004$. Dead χ_3 (five rungs) and dead χ_4 -20: $\text{ind}_- A_0 \in \{0, 1\}$ on all of them. Living χ_3 37/37 and living χ_4 41/41 likewise. P1 is not world-separating. The death of dead χ sits at $\text{sch} > 0$ in the r401 chart, i.e. entirely in the edge.

Machine check

Standalone: Fractions frame, omit- ℓ , rank-one interlacing, Weyl, threshold $\times 2$. Construction: FRAME-A lift rank, orthogonality, source-frame failure, scramble 21, two-period, dead/living χ , phase-blindness. Discovery census: `bulk_one_defect_probe.py` (`--smoke` / `full`), core-42, Twin, living and dead χ . Exit line: BULK ONE DEFECT VERIFIED.

Claim boundary

Finite identities for the frame implication, rank- r interlacing and Weyl; a named census that the bulk perturbation is a high-rank lift of a Bernstein–Szegő dual, not a rank-one defect, and that C_m -phases do not predict ind_- ; named scramble / two-period / mutant / dead- χ controls. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

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- [3] S. Hamann, *High-moment inertia*, standalone note, August 2026, `rh/problem/high_moment_inertia.tex`.
- [4] S. Hamann, *Source Weyl energy*, standalone note, August 2026, `rh/problem/source_weyl_energy.tex`.
- [5] S. Hamann, *The augmented edge signature*, standalone note, August 2026, `rh/problem/edge_signature.tex`.